

A histogram

A histogram uses adjoining bars to show frequencies or relative frequencies in numerical intervals, also called classes or bins.

How does its horizontal axis differ from a categorical bar graph?

Define the intervals

Synthetic download times: 2, 4, 5, 7, 9, 10, 13, 18 seconds. Use intervals $[0,5)$, $[5,10)$, $[10,15)$, $[15,20)$.

Where does a time of exactly 5 seconds belong?

Class width and boundaries

Class boundaries mark interval edges. Class width is upper boundary minus lower boundary. Synthetic whole-number counts range from 2 through 13.

Starting at 1.5, what width and boundaries give four equal classes covering all the values?

Precision and boundaries

Synthetic measurements are recorded to one decimal place, with minimum 2.3 seconds.

What starting boundary, one decimal place more precise, avoids putting 2.3 on an edge?

Count the bins

Synthetic download times: 2, 4, 5, 7, 9, 10, 13, 18 seconds. Use intervals $[0,5)$, $[5,10)$, $[10,15)$, $[15,20)$.

What frequency belongs in each interval?

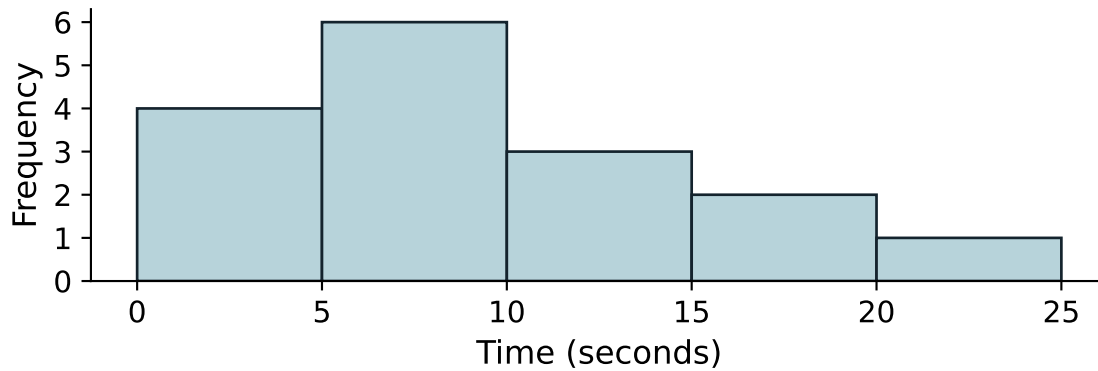
Constructing a histogram

Synthetic counts: 1, 1, 2, 2, 2, 3, 4, 4. Use boundaries 0.5, 1.5, 2.5, 3.5, 4.5.

How would you draw and label the frequency histogram?

Read a histogram

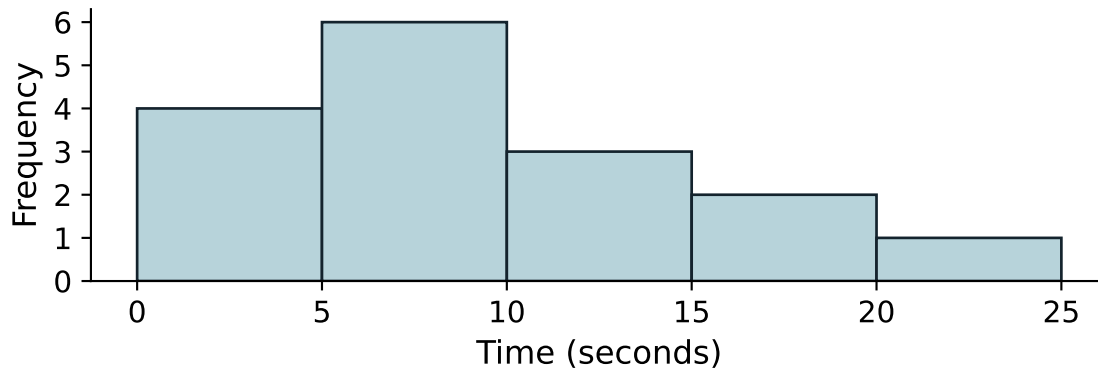
Synthetic times; bins are left-inclusive.



How many downloads took less than 10 seconds?

Read an interval

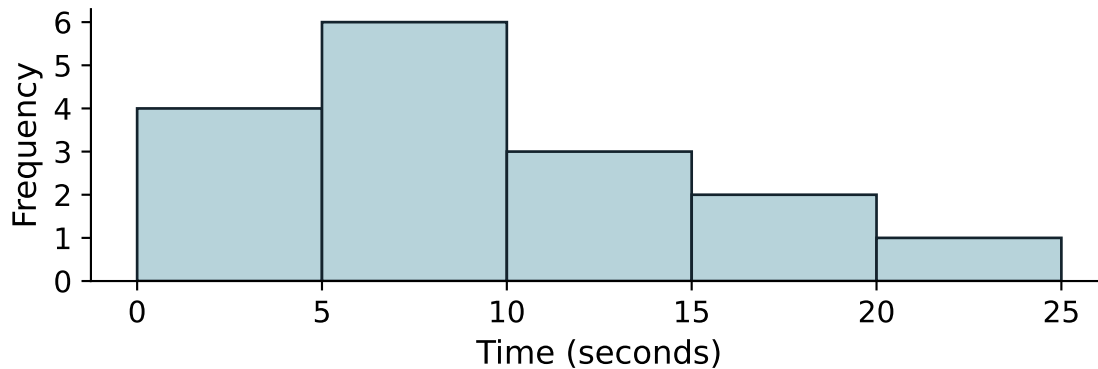
Synthetic times; bins are left-inclusive.



Which five-second interval contains the most observations?

Frequency to percentage

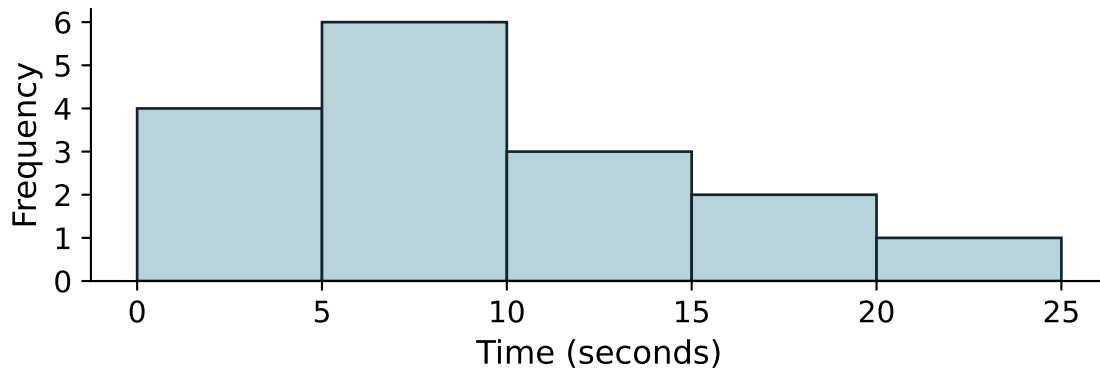
Synthetic times; bins are left-inclusive.



What percentage took from 10 to less than 15 seconds?

Information hidden by grouping

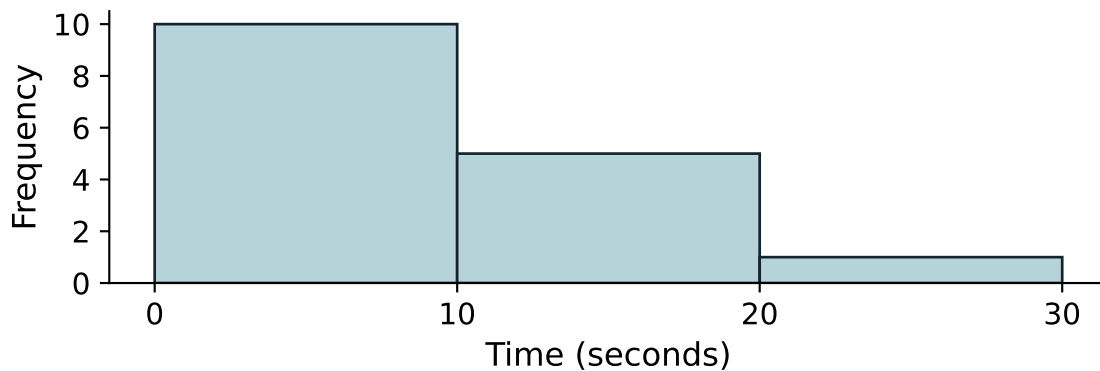
Synthetic times; bins are left-inclusive.



Can this graph reveal the exact number of times equal to 8 seconds?

Bin width

Synthetic times; bins are left-inclusive.



How does a wider bin change the detail we can see?

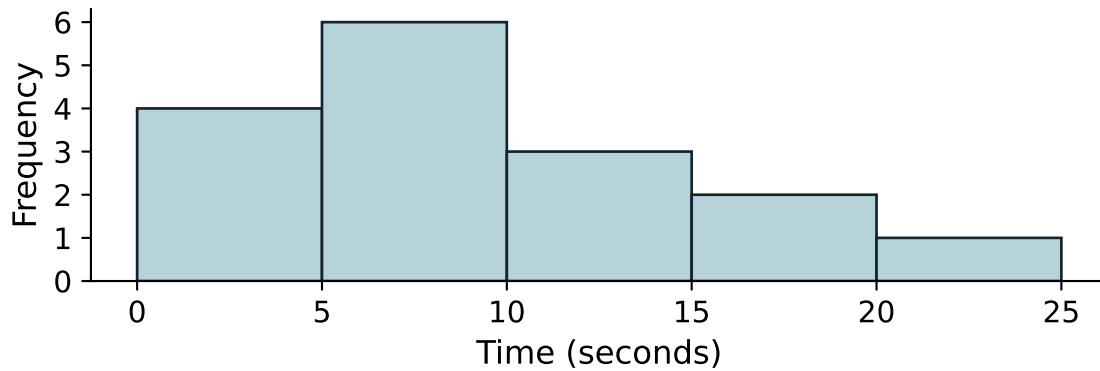
A gap

Synthetic file sizes (KB): 11, 12, 14, 16, 31, 33, 35.

What will a histogram with bins $[10,20)$, $[20,30)$, $[30,40)$ reveal?

Describe the shape

Synthetic times; bins are left-inclusive.



Is this distribution symmetric or skewed (asymmetric)?

Label the axes

A histogram has bars of heights 4, 7, and 9 but no axis labels.

What must we know before interpreting these bars?

Relative-frequency histogram

Synthetic intervals have counts 4, 7, 9 out of 20 files.

What heights would a relative-frequency histogram use?

A frequency polygon

A frequency polygon joins frequencies plotted at class midpoints.

For intervals $[0,5)$, $[5,10)$, $[10,15)$, where are the midpoints?

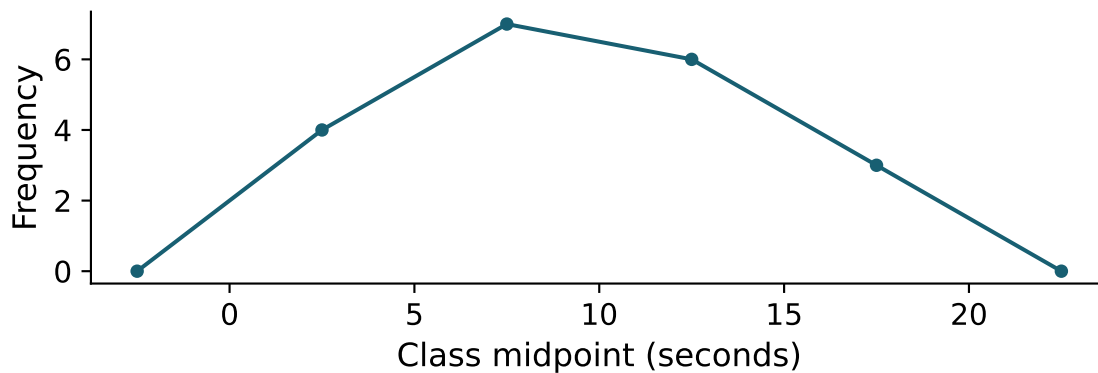
Build a frequency polygon

Synthetic download times. Equal-width class midpoints are 2.5, 7.5, 12.5, 17.5; counts are 4, 7, 6, 3.

What points should the polygon connect?

Read a frequency polygon

Synthetic grouped times; zero endpoints close the polygon.



Which class has the greatest frequency?

Comparing frequency polygons

Synthetic equal-sized groups use midpoints 5, 15, 25. A has frequencies 8, 6, 2; B has 2, 6, 8.

How would overlaying the two polygons show the shift between groups?

A time series

A time series records a variable in chronological order.

Why must the horizontal axis preserve time order?

Paired data sets

In a paired data set, each value of one variable matches one value of another. A time series pairs times with measurements.

What goes wrong if daily totals are sorted without keeping their dates attached?

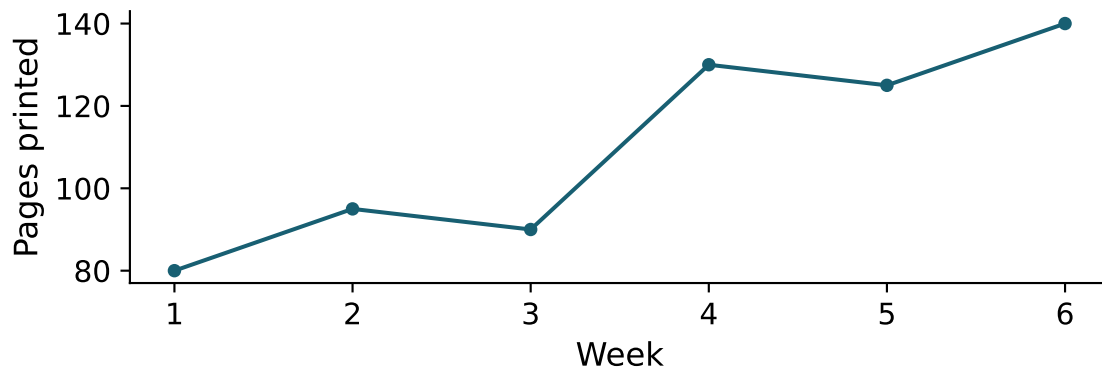
Constructing a time series graph

Synthetic daily print totals: Monday 20, Tuesday 35, Wednesday 25, Thursday 40.

How would you draw a labeled graph that preserves chronological order?

Read a time series

Synthetic weekly print totals.



During which adjacent weeks did printing increase most?

Distribution or time order?

Synthetic weekly print totals: 80, 95, 90, 130, 125, 140.

What information would a histogram lose compared with a time-series graph?

Unequal widths

One histogram bin spans 0–5 seconds; another spans 5–20 seconds. Both contain 10 observations.

Why might equal bar heights misrepresent the concentration of data?

Choose a graph

We record file sizes and the total files downloaded on each of 30 consecutive days.

Which graph would best show the change in daily totals?

Misleading vertical scales

Synthetic weekly totals rise from 98 to 102. A graph shows only the vertical range 97 to 103.

How could this scale exaggerate the apparent change?

Misleading time scales

A graph uses equal horizontal spacing for 2020, 2025, and 2026.

What misleading impression could unequal time intervals create?

Comparing pie charts

Synthetic totals: last year 100 purchases, 40% digital; this year 200 purchases, 30% digital.

Did the number of digital purchases fall just because its pie-chart share fell?